

Alpha Protocol Specification

Section 1: Introduction

This document describes the Alpha Protocol for secure data transmission. The Alpha Protocol was first published in 2019 and has been revised three times since then.

Section 2: Security Requirements

All data transmissions using the Alpha Protocol must be encrypted with AES-256. The encryption key must be rotated every 90 days. Each rotation requires generating a new key pair and securely distributing the public key to all authorized recipients.

Section 3: Authentication

Authentication proceeds in two steps. First, the client presents a signed token issued by the identity provider. Second, the server validates the token against the current key registry. If the token is valid and the key has not been revoked, access is granted.

Section 4: Performance Characteristics

The Alpha Protocol achieves a throughput of approximately 10,000 transactions per second on standard hardware. Latency under load is typically under 50 milliseconds for authenticated requests.

Section 5: Limitations

The protocol does not support multicast transmission. It also cannot be used over connections that do not provide a reliable transport layer.

Section 6: Revision History

Version 1.0 - 2019-03-15: Initial publication.

Version 2.0 - 2021-06-01: Added key rotation requirements.

Version 3.0 - 2023-11-20: Added authentication step two.

Version 4.0 - 2025-02-10: Current version, added performance benchmark requirements.